



WS4 – Dynamic Simulations of Neuromechanics

The Good Old Days



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WS4: Dynamic Simulations of Neuromechanics

DAY 1

From neuron to muscle excitation

- Intro to modelling (10m)
- LIF neuron model (10 m)
- From spikes to muscle excitation exercises (1h)
- Q&A

DAY 2

Forward Sim + Control

- MuJoCo and forward modelling basics (25 m)
- Control basics and introducing programming challenge (15 m)
- Arm prototyping exercise (1h)
- Q&A

DAY 3

Advanced use-cases

- Reinforcement learning & motion synthesis (10 m)
- Simulations for vision-based robotics (15 m)
- Virtual prostheses, sim2real concerns and methods (20 m)
- Ankle exo challenge (1h)
- Q&A + buffer

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What kind of models?

We will model how spiking neurons generate muscle forces that drive controlled motion in articulated rigid bodies, like robots, or you!!



Why make models?

**You want to understand a problem.
“I know, I’ll simulate it.”
Now you have two problems.**

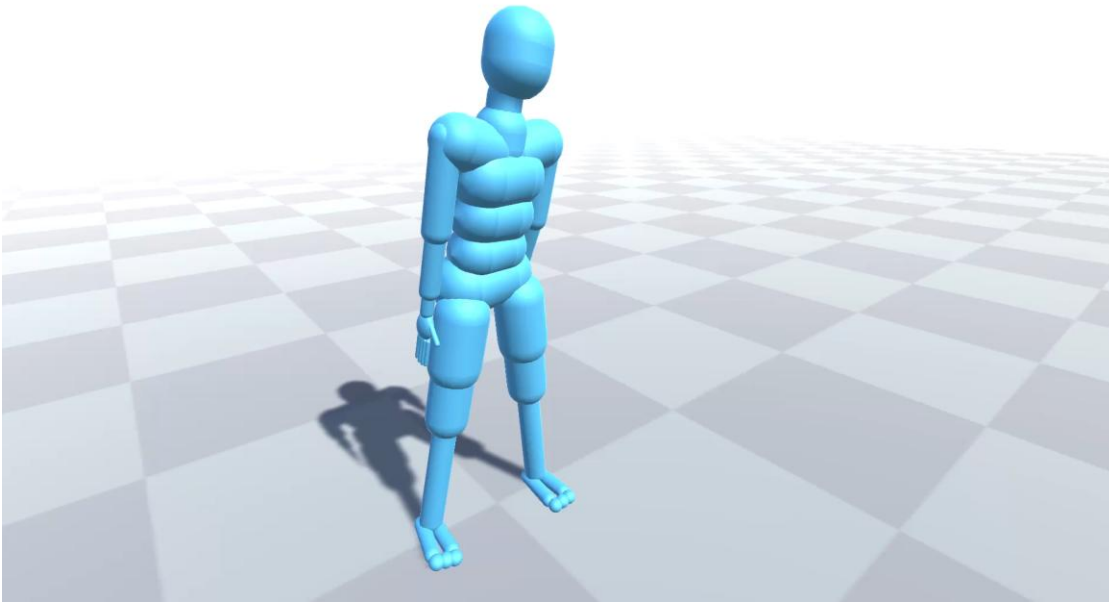


Why make models?



- **Test understanding**
- **Design and prototype**
- **Learn and transfer insights**

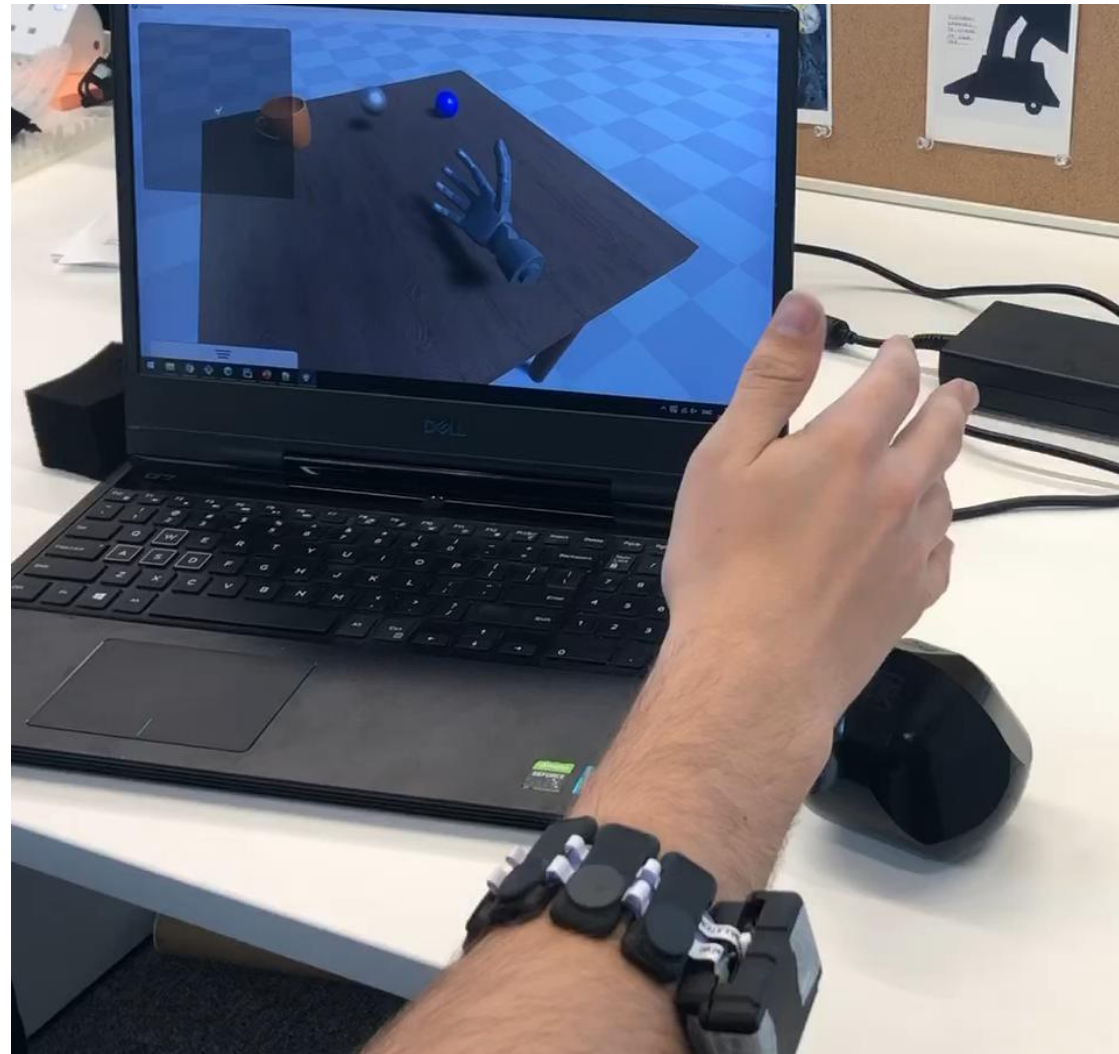
Simulate to test understanding



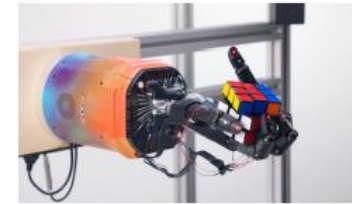
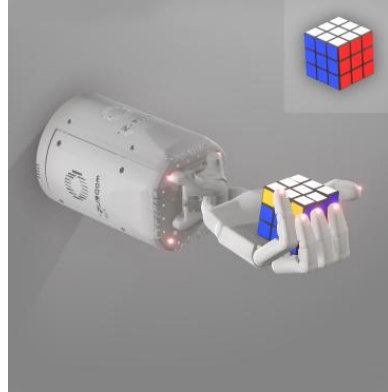
Pathological Gaits

Park, J., Min, S., Chang, P.S., Lee, J., Park, M.S. and Lee, J., 2022, July. Generative gaitnet. In *ACM SIGGRAPH 2022 Conference Proceedings* (pp. 1-9).

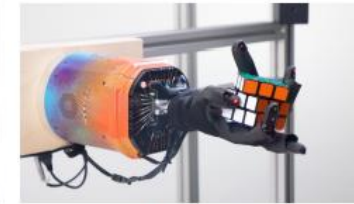
Simulate to prototype controllers and devices



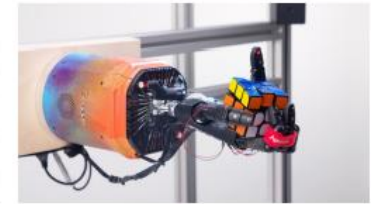
Simulate to learn



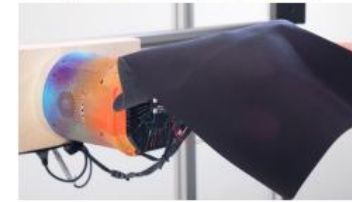
(a) Unperturbed (for reference).



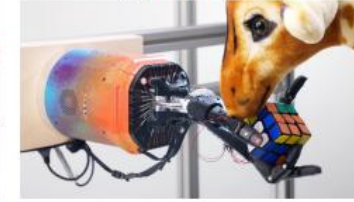
(b) Rubber glove.



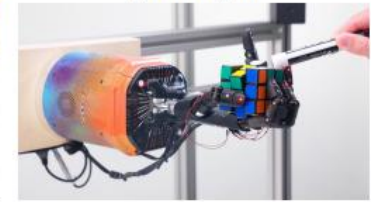
(c) Tied fingers.



(d) Blanket occlusion and perturbation.



(e) Plush giraffe perturbation.¹⁷



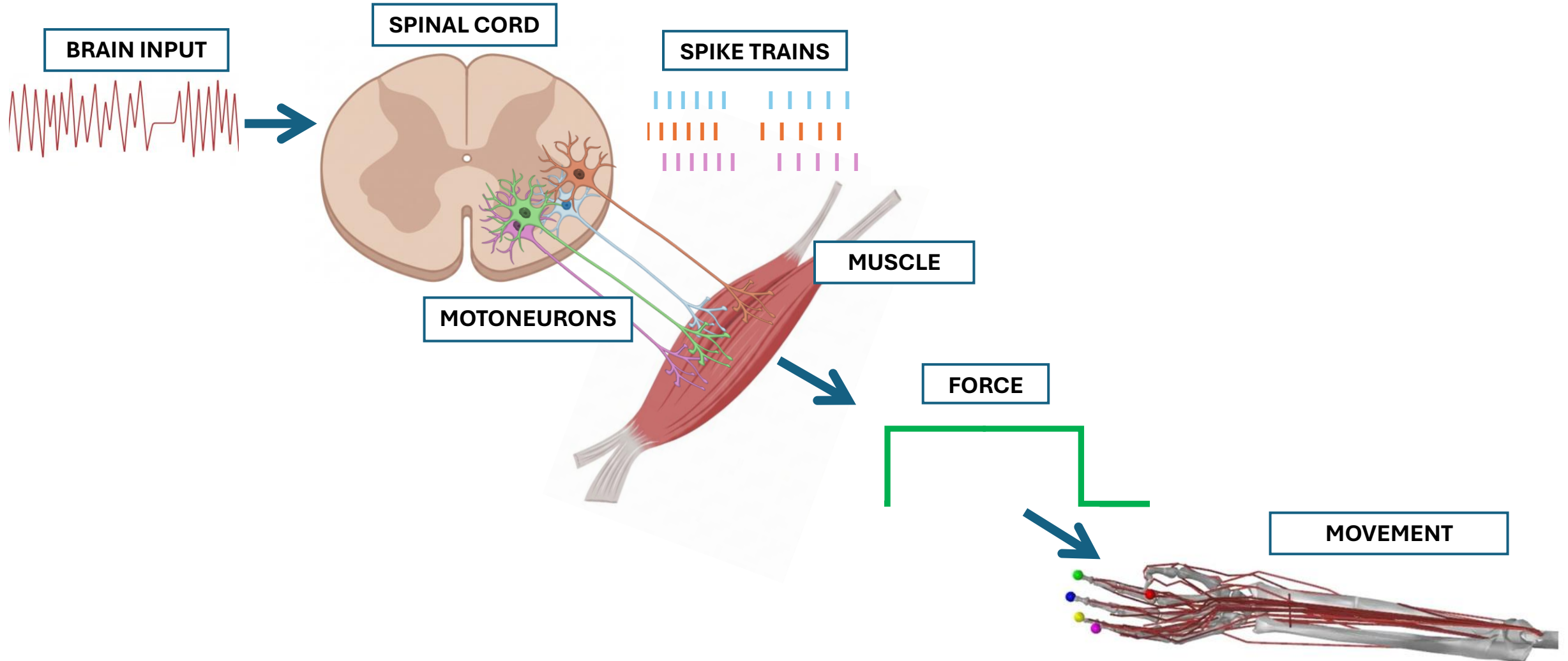
(f) Pen perturbation.

Akkaya, I., Andrychowicz, M., Chociej, M., Litwin, M., McGrew, B., Petron, A., Paino, A., Plappert, M., Powell, G., Ribas, R. and Schneider, J., 2019. Solving rubik's cube with a robot hand.

Hodossy, B. and Farina, D., 2022. Shared Autonomy Locomotion Synthesis with a Virtual Powered Prosthetic Ankle.



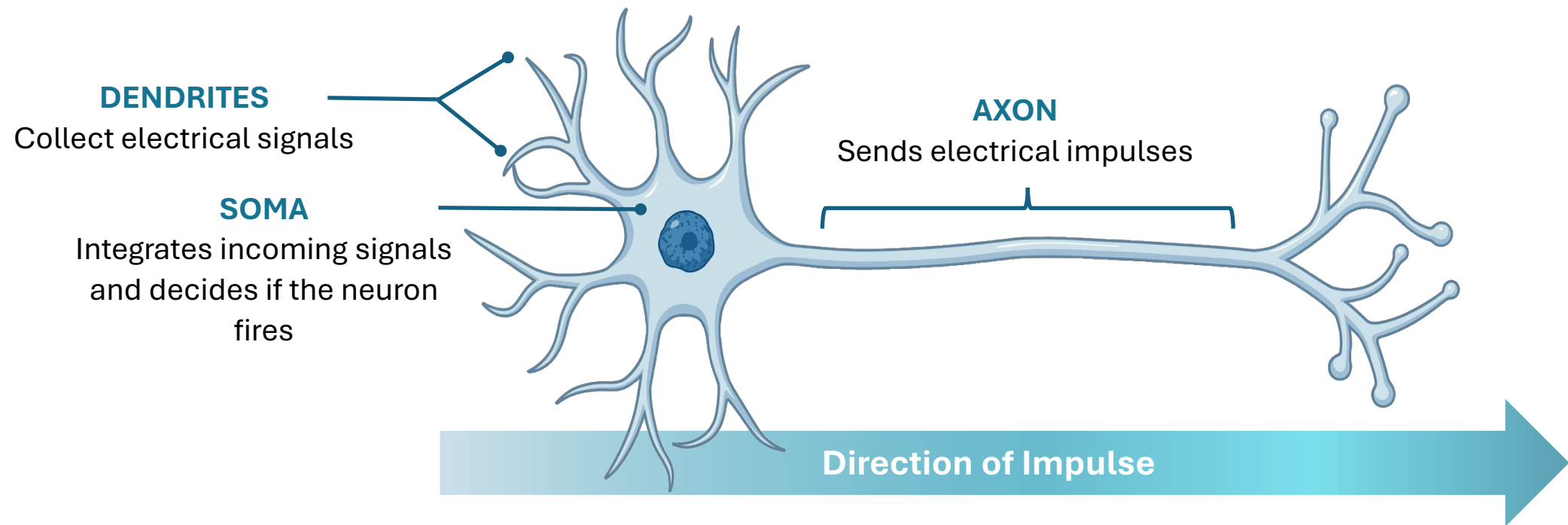
From Neuron to Movement



Hodossy & Farina, 2026, arXiv:2603.07364

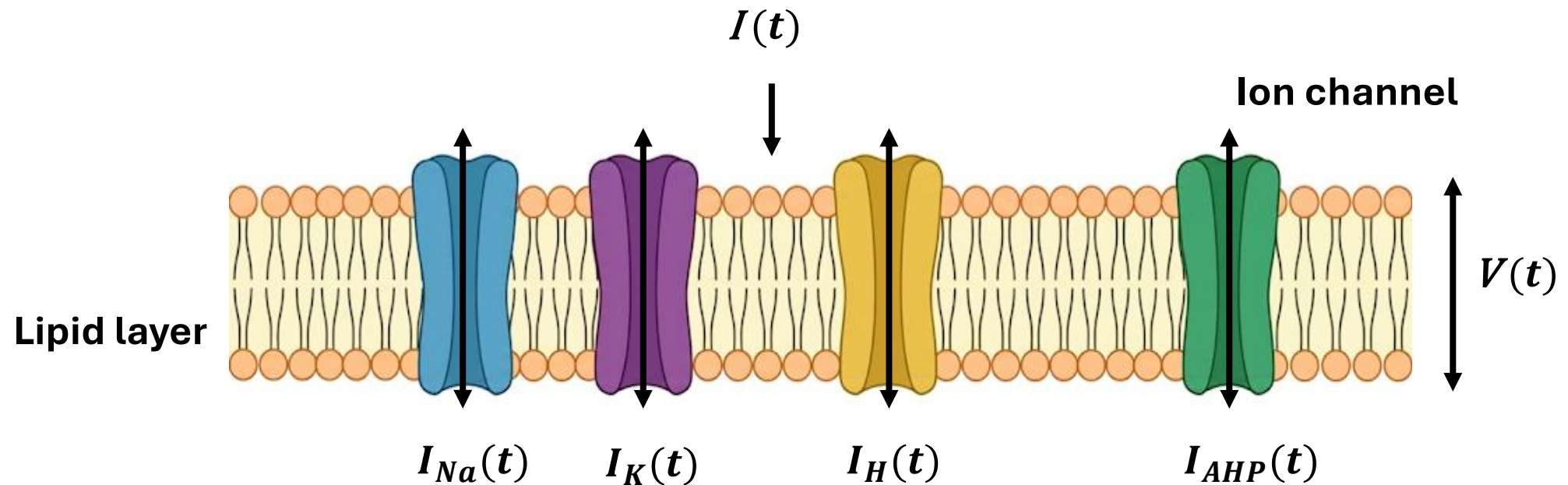
What is a neuron?

A neuron is a nervous system cell that **receives and transmits electrical signals**, while a **motoneuron** connects the **nervous system to muscles** and controls movement.

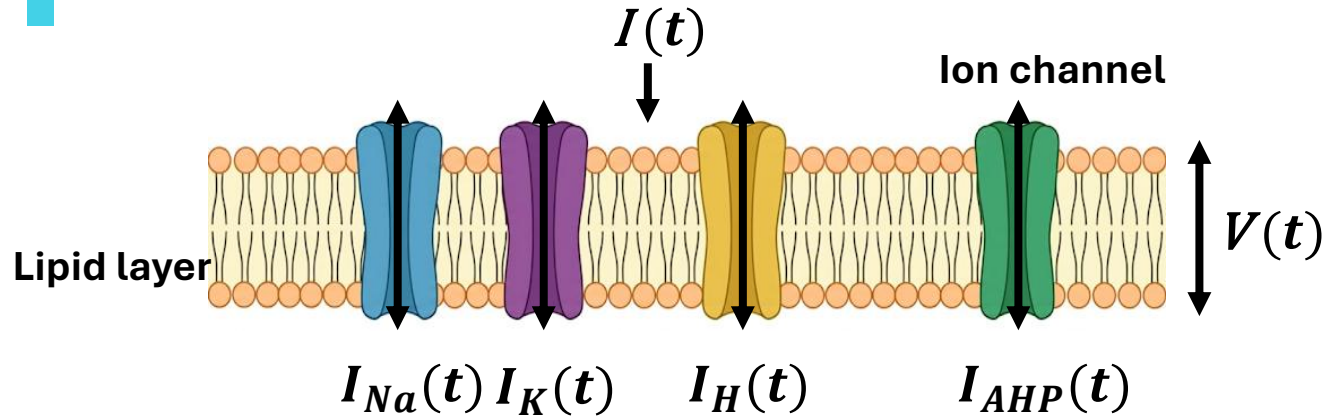


What is a neuron?

- $V(t)$: membrane potential
- $I(t)$: current on excitable membrane

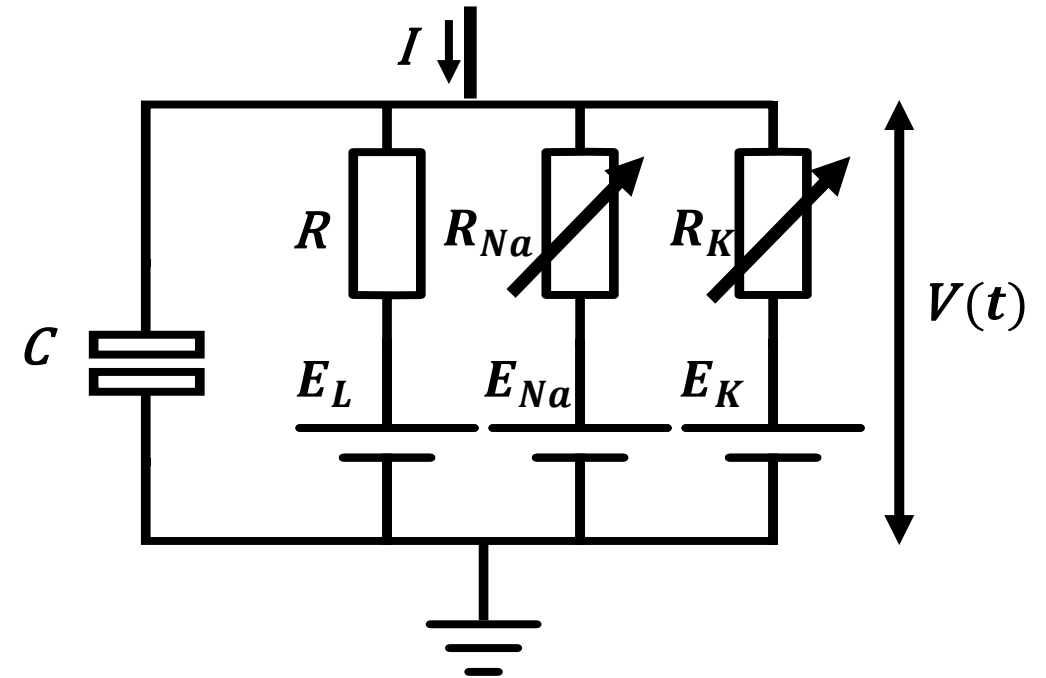


What is a neuron?



- $V(t)$: membrane potential
- $I(t)$: current on excitable membrane
- C : capacitance of the membra
- E_{Na}, E_K, E_L : Nernst potentials
- $\frac{1}{R} = g$: conductance ($Na, K, membrane$)

$$I(t) = C \frac{dV}{dt} + g_L(V - E_L) + I_{Na}(t) + I_K(t) + I_H(t) + I_{AHP}(t)$$



$$I(t) = C \frac{dV}{dt} + g_{Na}(V - E_{Na}) + g_K(V - E_K) + g_L(V - E_L)$$

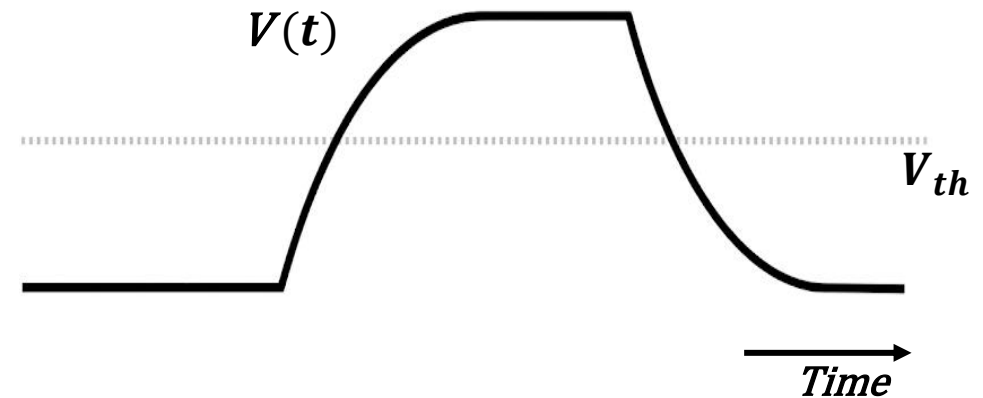
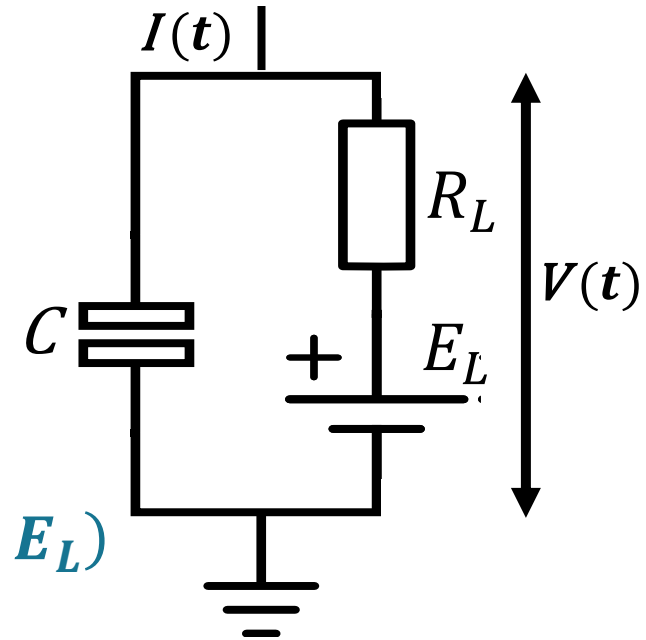
Leaky Integrate-and-Fire (LIF)

- $V(t)$: membrane potential
- $I(t)$: current on excitable membrane
- C : capacitance of the membrane
- E_L : Nernst potentials
- $\frac{1}{R} = g$: conductance (membrane)

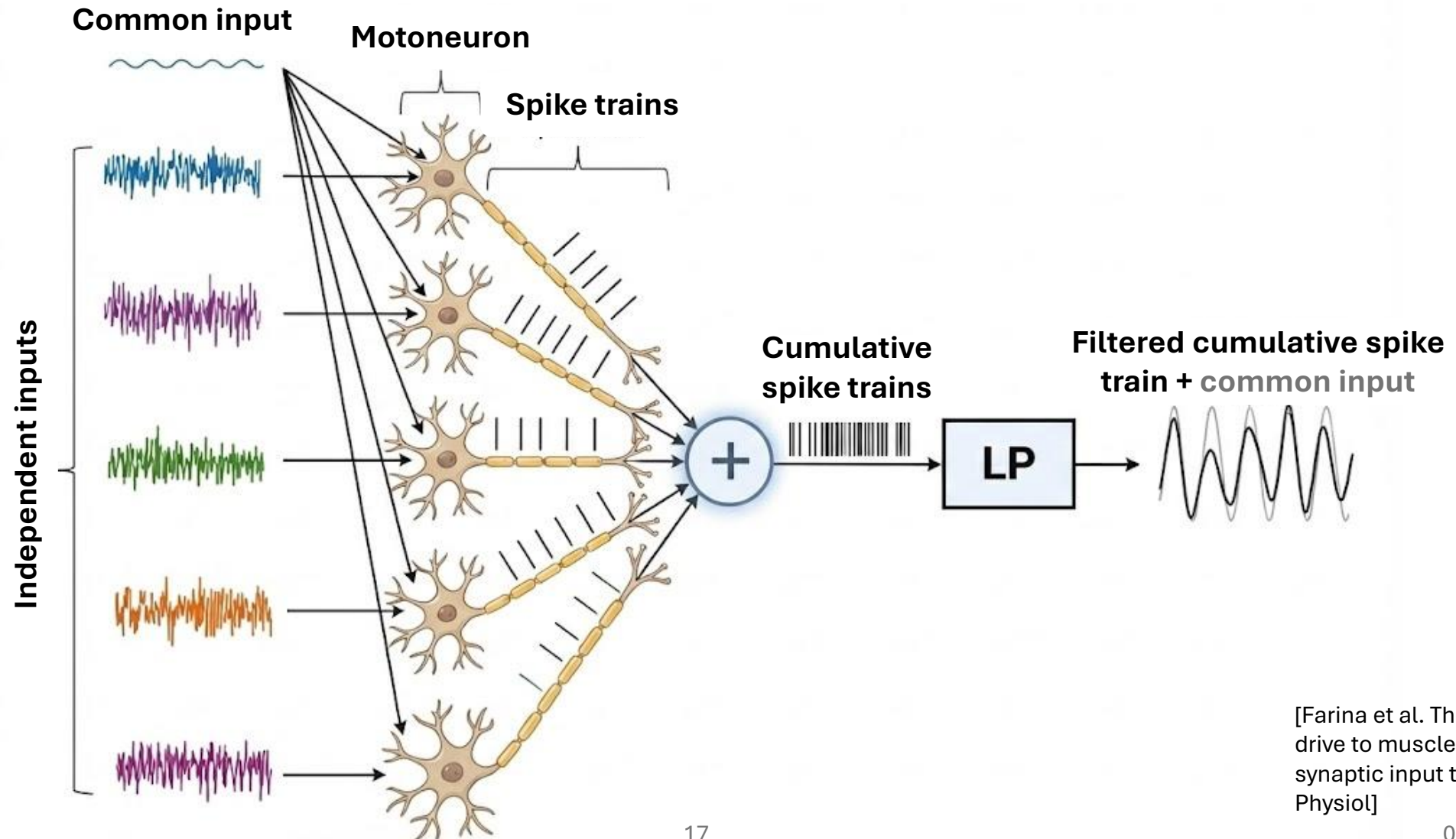
$$I(t) = C \frac{dV}{dt} + g_{Na}(V - E_{Na}) + g_K(V - E_K) + g_L(V - E_L)$$

$$C \frac{dV}{dt} = I(t) - g_L(V - E_L)$$

$$\text{If } V \geq V_{th} \text{ then } V(t + \Delta) = E_L$$



Neuron pool



[Farina et al. The effective neural drive to muscles is the common synaptic input to motor neurons. J Physiol]



Time to coding!!

Feel free to reach out to us if you have questions!

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Exercise materials:

https://github.com/Balint-H/ssnr_sim



Thanks!
Now it's beach time.